

FACULTY OF ENGINEERING

B.E.(CSE/CME/EEE) II - Semester (AICTE) (Main) (New) Examination, July/August-2025

Subject: Engineering Physics

Time: 3 Hours

Max. Marks: 70

Note: (i) First question is compulsory and answer any four questions from the remaining six questions. Each question carries 14 Marks.

(ii) Answer to each question must be written at one place only and in the same order as they occur in the question paper.

(iii) Missing data, if any, may be suitably assumed.

1. a) Explain the significance of population inversion in Lasers.
b) Define Hall effect.
c) What is Meissner effect? Explain.
d) What is Qubit? Explain.
e) Explain the significance of surface to volume ratio.
f) What is Displacement current? Explain.
g) What do you mean by Magnetostriction.
2. a) Explain the working of the Laser Induced Breakdown Spectroscopy (LIBS).
b) What are Ultrasonic and Infrasonic waves. Discuss Ultrasonic Pulse–Echo Testing Method.
3. a) Define Direct and In-direct Band gap semiconductors. Discuss the concept of Harvesting Energy Devices.
b) Mention Maxwell's equation in Integral form. State and prove Poynting theorem.
4. a) Explain Weiss Molecular field theory. Discuss about Hysteresis loop.
b) Discuss about High-Tc Superconductor and write their Industrial applications.
5. a) What is wave function? Discuss about Energy of particle in 1-D Potential Box.
b) Comment on No-Cloning Theorem. Mention applications of Quantum Computing.
6. a) Describe How Scanning Electron Microscope (SEM) is used in material Characterization.
b) Describe Electron Beam-Evaporation method of thin film preparation.
7. a) Obtain an expression for Time-Independent Schrodinger wave equation.
b) Discuss Kronig-Penny model in detail.
